Exam 2 Review

ENGR 1411 – Material Science

Exam #2 Format

Required Section:

- Millers Indices
  - Points
  - Directions
  - Planes
  - LD, PD calcs
- General Knowledge
  - Multiple choice, fill in the blank, true/false, etc.

Proficiency Retakes:

- **P1: Units**
- **P2: Stress-Strain Curve**
- **P3: Stress-Stain Calcs**

What does the exam cover?

Lectures + In class exercises:

09 – Impact and Fatigue

Time Dependent Strain

10 – Crystallographic Directions

11 – Deformation and Slip Systems

Fracture

12 - Strengthening and Toughening

Lab Modules

04 – DBTT - Toughness

05 – CW and Heat Treatment (prelab - coldworking)

Homework: HW7 – HW10

You are allowed

- Calculator
- Pencil
- You will be provided the equation sheet
- An 8.5” x 11” sheet of handwritten notes is allowed in the exam
  - These will be collected with your exam.
  - No solved example problems. Solved example problems on your equation sheet will result in 0 pts for similar problems on the exam.
  - Can reuse previous notesheet

Part 1: General Knowledge

<u>Question Types</u>: Multiple choice, fill in the blank, true/false, etc.

<u>Examples</u>: In-class exercises, past quizzes

Can come from all lectures and modules:

- 09 – Impact and Fatigue, Time Dependent Strain
- 10 – Crystallographic Directions
- 11 – Slip systems and Fracture
- 12 – Strengthening and Toughening
- Module 4 and Module 5 (prelab only – cold working)

Part 2: Millers Indices

Question Types: Given direction/plane, find index. Given index, draw direction or plane. Calculate Linear Density, Planar Density

Examples Problems:

- HW 8
- 10 – CE Crystallographic Directions and Planes
- 11 – CE Deformation p7-8

Lectures/Lab:

- 10 – Crystallographic Directions and Planes
- 11 – Deformation and Slip Systems

Once you finish the required portion (new content), you may test on proficiency exams

P1: Units

<u>Question Type:</u>

- Report units for variables in SI and US systems
- Convert units from SI to US, nm to mm, etc

<u>Examples Problems</u>:

  – HW6 – Unit questions
  – P1 Unit Test review questions

<u>Lectures/Lab</u>:

- 07 - Units

P2: Stress/Strain Curve

Question Type: Given a stress strain curve, determine the parameters

Examples Problems:

- HW 5 – Properties practice Question 1 and 2
- 5 – CE – Stress Strain Curve (p. 13 – 23)
- P2 – review test problems

Lectures:

- 05 – The Stress Strain Curve
- 06 – More Elastic properties (review slides)

P3: Stress Strain Calculations

Question Type: Open ended calculations.

Examples Problems:

- 5 – CE – Stress Strain_Curve (p. 14)
- 6 – CE – More Elastic Properties
- HW 6 – Units and Stress Calcs
- P3 test review problems

Lectures/Lab:

- 05 – The Stress Strain Curve
- 06 - More Elastic Properties
- 07 – Units

Exam 2 Study Session

- Lauren will be running another student session before the exam
- Monday 8-9 PM
- JLK 323

Questions?

For the rest of class time:

- Work on Miller indices practice.
- Ask questions about proficiency tests
- Work through practice material
- Start creating your 1 page note sheet.